The Infant Who Spits: Happy Spitters, Crying That Is Not Reflux, and the Red Flags That Point Somewhere Else.

Picture a ten-week-old girl brought to clinic by tired parents. She brings up milk after most feeds, sometimes a mouthful, sometimes what looks like the whole feed across her father's shoulder. In the evenings she cries for hours, arches her back and pulls away from the bottle. A relative has told them it is reflux, and an online forum has told them which medicine to ask for. On the growth chart her weight runs comfortably along the fiftieth centile. Her fontanelle is soft, her abdomen is soft, and between the episodes she smiles at you.

The clinical question of this chapter is which regurgitating infants need nothing more than an explanation, and which ones are vomiting for a reason that has nothing to do with the oesophagus. The main idea is that the diagnosis of physiological regurgitation is made by the history, the examination and the growth chart, and that it is safe only when the warning signs have been asked about deliberately, one by one. The job of the visit is to find the small number of infants whose vomiting comes from obstruction, from the brain, from infection, from a metabolic disease or from allergy, and to protect everyone else from tests and drugs they do not need. Two numbers frame the stakes. In one surgical series, when malrotation of the gut presented with bilious vomiting, eighty percent of infants already had a volvulus, a twist of the bowel on its blood supply, at operation. At the other end of the spectrum, in a randomised trial of crying infants thought to have reflux, a proton pump inhibitor helped exactly as many babies as a placebo did, fifty-four percent in each group, and caused more serious side effects.

A note on sources. The main document is the 2018 joint guideline on gastro-oesophageal reflux in infants and children from the North American Society for Pediatric Gastroenterology, Hepatology and Nutrition, known as NASPGHAN, and the European Society for Paediatric Gastroenterology, Hepatology and Nutrition, known as ESPGHAN, written by Rosen and colleagues. Alongside it sit the British guideline from the National Institute for Health and Care Excellence, known as NICE, numbered NG1, published in 2015 and updated in 2019, and the Rome four criteria for functional gastrointestinal disorders of infants and toddlers, published by Benninga and colleagues in 2016. The evidence on crying comes from Heine and colleagues in 1995, Moore and colleagues in 2003, Orenstein and colleagues in 2009 and a pooled analysis of crying diaries by Wolke and colleagues in 2017. The evidence on questionnaires comes from Salvatore and colleagues in 2005, Van Howe and Storms in 2010 and van Lennep and colleagues in 2024. The warning signs draw on surgical series of bilious vomiting, a large study of antibiotics and pyloric stenosis by Eberly and colleagues in 2015, and the European birth cohort study of cow's milk allergy called EuroPrevall.

Part one. Not everything that comes up is reflux.

Start by separating two different events. Gastro-oesophageal reflux, which we shorten to GER, is the passage of stomach contents into the oesophagus, with or without regurgitation into the mouth. In infants it is driven by transient relaxations of the lower oesophageal sphincter, the muscular valve at the bottom of the gullet, which open for ten seconds or more without a swallow. A liquid diet, a small stomach, large feeds relative to body size and much time spent lying down make those relaxations more likely to carry milk upward. Regurgitation is therefore passive and effortless. It usually happens soon after a feed in an infant who is otherwise comfortable.

Vomiting is different. It is an active reflex, coordinated by the brainstem, with retching, forceful expulsion, pallor and distress. The trigger for that reflex can come from almost anywhere. It can come from the outlet of the stomach or the bowel, as in pyloric stenosis, malrotation, intestinal atresia or intussusception. It can come from the brain, as in hydrocephalus, intracranial haemorrhage, a tumour or abusive head trauma. It can come from infection, such as a urinary tract infection, sepsis, meningitis or gastroenteritis. And it can come from metabolism or allergy, such as a urea cycle disorder, galactosaemia, adrenal crisis, cow's milk protein allergy or eosinophilic disease. Taking a history that separates effortless regurgitation from true vomiting is the first diagnostic test of the visit, and it costs nothing.

The formal definitions follow the same logic. The 2018 guideline defines GER as the passage of gastric contents into the oesophagus with or without regurgitation and vomiting, and defines gastro-oesophageal reflux disease, or GERD, as reflux that leads to troublesome symptoms affecting daily functioning, or to complications. Regurgitation on its own is not a disease.

Part two. The happy spitter.

The infant who regurgitates often but is thriving and content has earned an affectionate name, the happy spitter. The NICE guideline describes this pattern with useful numbers. At least forty percent of infants regurgitate effortlessly. It usually begins before eight weeks of age. About five percent of infants have six or more episodes a day, so frequency alone does not make disease. And about ninety percent resolve before the first birthday. The Rome four document adds that between forty-one and sixty-seven percent of healthy four-month-olds regurgitate at least once a day. The classic American survey by Nelson and colleagues, published in 1997, asked nine hundred and forty-eight parents and found daily regurgitation in about half of infants up to three months old, a peak of sixty-seven percent at four months, and only five percent at ten to twelve months. Even at the peak, at most twenty-three percent of parents called it a problem. The 2018 guideline adds a point about timing that becomes important later: physiological reflux seldom starts before one week of age or after six months.

Rome four gives a formal definition of infant regurgitation. The infant must be between three weeks and twelve months old, must regurgitate twice or more a day for at least three weeks, and must be otherwise healthy. Equally important are the features that must be absent: retching, vomiting blood, aspiration, apnoea, failure to thrive, difficulty feeding or swallowing, and

abnormal posturing. In other words, the definition contains its own list of warning signs. When parents bring a happy spitter, what they want is a reason and a plan. They are dealing with laundry, worry and advice from everyone around them. The treatment is a clear explanation of the natural history and of what would make you worried.

Part three. Crying is not reflux.

This is where most over-treatment begins, because infant crying is common, distressing and easy to attribute to the reflux that parents can see. The facts point the other way.

First, crying has its own clock. In a pooled analysis of twenty-eight diary studies including eight thousand six hundred and ninety infants, Wolke and colleagues found that between seventeen and twenty-five percent of infants met criteria for colic in the first six weeks of life, about eleven percent at eight to nine weeks, and less than one percent, zero point six percent, at ten to twelve weeks. Average fussing and crying time was about two hours a day in the early weeks and fell to about an hour a day by ten to twelve weeks. If you draw these numbers as a bar chart, you see a tall bar for the first six weeks, a much smaller one at two months and almost nothing at three months. Regurgitation, by contrast, peaks at four months. The two curves are out of step, which is the first hint that one does not cause the other. Rome four now defines infant colic clinically as recurrent and prolonged crying, fussing or irritability that starts and stops before five months of age, that has no obvious cause and that caregivers cannot prevent or resolve, in an infant without failure to thrive, fever or illness. The old rule of three hours a day on three days a week is kept only for research.

Second, crying does not track acid. Heine and colleagues studied seventy irritable infants referred with presumed reflux, all of whom had oesophageal pH monitoring, a test that measures acid in the gullet over twenty-four hours. Pathological acid reflux was found in only one of twenty-four infants under three months old, about four percent, compared with ten of forty-six older infants, about twenty-two percent. Every infant with pathological reflux also vomited frequently, so there was no silent reflux in this series. And the behaviours parents describe, such as back-arching, sleep disturbance, refusing feeds and poor weight gain, did not predict an abnormal study. Rome four puts it plainly: there is a lack of correlation between crying, irritability and reflux.

Third, removing acid does not stop crying. Moore and colleagues ran a crossover trial in thirty irritable infants aged three to twelve months, in which each infant received omeprazole and placebo in turn. Omeprazole did exactly what it is designed to do: the reflux index, the percentage of time the oesophagus is acidic, fell by about nine points on omeprazole and by about two points on placebo, a highly significant difference. But crying and fussing fell from about two hundred and sixty-seven minutes a day to about one hundred and eighty-eight minutes a day regardless of which treatment the infant was receiving. The probe improved; the baby did not. Time, not acid, settled the babies. Orenstein and colleagues then randomised one hundred and sixty-two infants with symptoms attributed to reflux to lansoprazole or placebo. Fifty-four

percent responded in each group. Serious adverse events, mainly lower respiratory tract infections, occurred in ten infants on lansoprazole against two on placebo. A 2015 meta-analysis of five randomised trials by Gieruszczak-Bia■ek and colleagues concluded that proton pump inhibitors do not reduce crying or irritability in infants. The 2018 guideline therefore recommends against using either histamine-two receptor antagonists or proton pump inhibitors to treat crying or distress in otherwise healthy infants, and against using them for visible regurgitation in otherwise healthy infants.

Part four. Questionnaires and labels.

It is tempting to use a questionnaire to decide which crying, spitting infant has reflux disease. The most widely used is the revised Infant Gastro-oesophageal Reflux Questionnaire, abbreviated I-GERQ-R. It has twelve items, scores from zero to forty-two, and is considered positive at sixteen or above. It was validated to follow symptoms over time, not to diagnose disease, and the numbers show why. In a study of unselected infants by Van Howe and Storms, about a quarter, twenty-five and a half percent, scored sixteen or more at one month of age, falling to under three percent at six months, because the crying items inflate the score in the colic weeks. In a 2024 Dutch study of nine hundred and seventy-nine healthy infants, sixteen percent of those under one month scored sixteen or more, and the authors concluded that such scores should not necessarily be considered abnormal. And when Salvatore and colleagues compared the questionnaire with biopsy and pH monitoring in a hundred infants with suspected reflux disease, the questionnaire was positive in eighty-one percent of infants whose biopsy and pH study were normal, and missed about a quarter of those who did have disease. Symptoms, histology and pH monitoring correlated poorly. A score is not a diagnosis.

The words we use matter as much as the tests. In an experiment by Scherer and colleagues, published in 2013, one hundred and seventy-five parents read a short description of a crying infant. Some were told the baby had a disease called gastro-oesophageal reflux disease, and some were not; some were told a medicine would work, and some that it probably would not. When the disease label was used, parents still wanted the medicine even after being told it was ineffective. Without the label, being told the drug did not work reduced their interest. The label sells the drug. And the drug has costs. Besides the infections seen in the Orenstein trial, a study by Malchodi and colleagues of eight hundred and fifty-one thousand six hundred and thirty-one children in the American military health system found that eleven percent had received acid suppression in their first year, and that fracture risk later in childhood was higher after proton pump inhibitors, more so when combined with histamine-two blockers, though not with histamine-two blockers alone. The better words are simple: spitting up, normal, expected to end by about twelve months, and here is what would make me want to see her again.

Part five. The red flags and where they point.

The 2018 guideline lists warning signs in an infant who regurgitates or vomits, and NICE attaches an action to many of them. It helps to hold each flag together with the diagnosis it points to.

Bilious vomiting, meaning green vomit, points to malrotation with volvulus, intestinal atresia or another obstruction, and calls for same-day surgical review and an upper gastrointestinal contrast study. Persistent, forceful or projectile vomiting in the first two months points to pyloric stenosis and calls for same-day assessment and ultrasound. Vomiting blood, once swallowed maternal blood has been excluded, points to oesophagitis, gastritis or a bleeding disorder. Onset after six months of age, or persistence beyond twelve to eighteen months, points away from physiological reflux and towards infection, particularly urinary infection, neurological disease or other gastrointestinal disease. Fever, lethargy or painful urination point to urinary infection, sepsis or meningitis. A bulging fontanelle, a rapidly growing head, seizures, or a head that is too large or too small point to raised intracranial pressure or structural brain disease and call for urgent neurological assessment and imaging. Abdominal distension, rectal bleeding or chronic diarrhoea point to obstruction, necrotising enterocolitis, cow's milk allergy or another enteropathy. Weight loss, faltering growth and vomiting at night point to any of these, or simply to inadequate intake, and should never be called reflux until proven. Excessive irritability or pain counts as a flag only together with other features; on its own, the guideline notes, it is unlikely to be related to reflux disease.

The colour green deserves particular attention. In older series, between thirty and forty percent of newborns with bilious vomiting needed an operation; in a more recent cohort of term infants, the figure was about fifteen percent, with malrotation in about three percent. The proportion may be falling, but the stakes are not. In the series of Seashore and Touloukian, thirty-five of forty-four infants with malrotation who presented with bilious vomiting, eighty percent, already had a volvulus at surgery. The abdomen can remain soft until the bowel is dead. NICE therefore asks for a same-day upper gastrointestinal contrast study for unexplained bile-stained vomiting, and this study had a sensitivity of ninety-six percent for surgically confirmed malrotation in one large radiology series. The second difficulty is recognising the colour. When Walker and colleagues showed a colour chart to parents and general practitioners, fifty-six percent of parents and twenty-five percent of general practitioners did not choose any green shade as bile. Ask whether the vomit was green like grass or spinach, and remember that yellow milk curd is not bile.

Timing is a diagnosis in itself. In the first week of life, physiological reflux rarely starts, so new vomiting should make you think of obstruction, metabolic disease or sepsis. Between about two and eight weeks lies the window for pyloric stenosis, typically in a hungry infant who vomits projectile, non-bilious feeds and wants to feed again immediately. Around four months, physiological regurgitation peaks at sixty-seven percent, by which time colic has already faded. New vomiting that begins after six months is not physiological reflux; think of urinary infection, neurological disease and, in the older infant, eosinophilic oesophagitis. And persistence beyond

twelve to eighteen months means that the natural history has failed, so re-examine the diagnosis and the anatomy. NICE adds a practical rule: test the urine if regurgitation comes with faltering growth, starts late, after eight weeks of age, or is frequent with marked distress. Rome four also describes a separate disorder, infant rumination, in which repeated effortless regurgitation lasting at least two months begins between three and eight months of age, often with self-stimulating behaviour; it is not reflux.

For pyloric stenosis, ask about antibiotics. Eberly and colleagues studied more than one million infants in the American military health system, of whom two thousand four hundred and sixty-six developed pyloric stenosis, about two point three per thousand. Exposure to erythromycin in the first fourteen days of life multiplied the odds about thirteen-fold, with an adjusted odds ratio of thirteen point three. Azithromycin in the same window carried an odds ratio of about eight. Between days fifteen and forty-two the odds ratios were about four for erythromycin and three for azithromycin, and after six weeks there was no association. The horizontal bar chart of these figures shows the risk shrinking with each later window of exposure. On ultrasound, a pyloric muscle thickness above about three millimetres and a channel length above about fifteen millimetres are the usual thresholds.

Cow's milk protein allergy sits in the middle ground, real but frequently over-called. In the EuroPrevall birth cohort, Schoemaker and colleagues enrolled twelve thousand and forty-nine newborns across nine European countries. Three hundred and fifty-eight infants were suspected of having cow's milk allergy, but only fifty-five were confirmed by a double-blind food challenge, an incidence of about half a percent overall and about one percent in the United Kingdom and the Netherlands. In other words, only about one suspected infant in seven had the allergy. Sixty-nine percent of confirmed infants tolerated cow's milk one year later. Think of allergy when regurgitation comes with eczema, blood or mucus in the stool, chronic diarrhoea or a strongly atopic family, or when it persists despite good feeding advice. The 2018 guideline suggests a two- to four-week trial of an extensively hydrolysed or amino acid-based formula, or elimination of cow's milk from the mother's diet if the infant is breast-fed, when non-drug measures have failed. The 2024 ESPGHAN position paper by Vandenplas and colleagues adds that the diagnosis requires re-challenge, because over-diagnosis is far more common than under-diagnosis.

Part six. What to do, and what not to order.

The North American and European algorithm for the regurgitating infant runs in five steps. First, take the history and examine the infant, asking about each red flag in turn, reviewing the feeding history and plotting growth; if any flag is present, investigate and refer. Second, if the infant is thriving with no flags, give reassurance and anticipatory guidance, with no tests and no acid suppression. Third, adjust feeds: avoid overfeeding, offer smaller and more frequent feeds where volumes are large for weight, use thickened feeds, and continue breastfeeding. Pooled trials show that thickened feeds reduce regurgitation by about one episode a day and vomiting by

slightly less, although their effect on crying is unclear. Fourth, try two to four weeks of an extensively hydrolysed or amino acid formula, or maternal elimination. Fifth, only then, and usually through a paediatric gastroenterologist, consider a four- to eight-week trial of acid suppression, and stop it if it does not help. Throughout, infants should sleep on their backs. Positioning for reflux is not advised because of the risk of sudden infant death.

Just as important is the list of tests not to order for the spitting infant. An upper gastrointestinal contrast series does not diagnose reflux disease, but it is the right test for anatomy, such as malrotation, and it is urgent for bilious vomiting. Abdominal ultrasound does not diagnose reflux either, but it is the test for pyloric stenosis. Gastric scintigraphy is strongly discouraged for reflux diagnosis. Oesophageal pH or pH-impedance monitoring is not for uncomplicated regurgitation; it is for correlating persistent troublesome symptoms with reflux, separating acid from non-acid reflux, or checking whether treatment is working. Endoscopy with biopsy does not diagnose reflux disease, but it is the test for complications, eosinophilic oesophagitis and vomiting blood. An empiric trial of a proton pump inhibitor is not a diagnostic test in infants, because five infant trials showed no benefit over placebo. And salivary pepsin and other biomarkers have no place in routine care. Even when pH-impedance is used, parents fail to record more than half of symptom events, which limits every symptom-association index, and the guideline does not prefer any of them.

Part seven. Where the guidelines agree and differ.

The North American and European guideline, NICE and Rome four agree on the core message. The healthy spitter needs reassurance, not investigation or drugs. Acid suppression should not be used for regurgitation alone. Bilious vomiting is a red flag, and NICE makes the action explicit with a same-day contrast study. They differ in several places. For crying or distress, the 2018 guideline recommends against acid suppression, whereas NICE allows a four-week trial when overt regurgitation comes with distress, feeding difficulty or faltering growth in an infant who cannot report symptoms. For urinary infection, NICE is materially more specific, asking for urine testing in late-onset regurgitation, faltering growth or marked distress. The 2018 guideline includes a hydrolysed formula trial, whereas NICE does not, and reaches allergy through its red-flag route. And for alginates, the 2018 guideline advises against chronic use, while NICE allows a one- to two-week trial in formula-fed infants after thickening. Neither document gives a number for overfeeding, which remains a gap; both simply say to avoid feeding volumes that are excessive for weight.

Pearls and pitfalls.

Four traps recur. The first is to equate crying with reflux; colic peaks in the first weeks, acid does not track crying, and the drug equals placebo with more infections. The second is to accept yellow vomit as bile, or to dismiss green vomit in a well-looking baby; green is surgical and needs a same-day contrast study. The third is to call new vomiting after six months late reflux; it

is not, so test the urine and examine the head and fontanelle. The fourth is to call regurgitation with vomited blood or blood in the stool just reflux; that belongs on an allergy, enteropathy or surgical pathway. And one more: starting an acid suppressor to see if it helps. Improvement over two weeks is the natural history, and once the label and the drug have been given, they are hard to take away.

Let us return to the girl from the beginning and then to some others. The ten-week-old who spits after most feeds, grows along the fiftieth centile and is content between episodes has physiological regurgitation; explain the natural history and order no tests and no drugs. Her evening crying is a colic pattern, so explain it, support the parents and review the feeding, without acid suppression. A four-week-old boy with projectile vomiting who is hungry afterwards and received erythromycin at ten days old has pyloric stenosis until proven otherwise; check electrolytes, arrange an ultrasound and obtain a same-day surgical review. A three-week-old with two green vomits and a soft abdomen, who looks well, needs a same-day upper gastrointestinal contrast study and a surgical opinion. A seven-month-old with a week of new vomiting and irritability, without fever, does not have reflux; send a urine culture and measure the head circumference, feel the fontanelle and examine the nervous system. A three-month-old with regurgitation, eczema, mucus in the stools and slow weight gain should have a two- to four-week trial of extensively hydrolysed formula followed by a planned re-challenge. And a thriving five-month-old who has been on omeprazole for three months for reflux should stop the drug, with an explanation of the natural history and a review in four weeks.

What to do differently in clinic.

So, in clinic tomorrow, let the growth chart and the red flags make the diagnosis. Ask about each warning sign out loud: green vomit, projectile vomiting, blood, fever, onset after six months, the fontanelle and the head circumference, diarrhoea and rectal bleeding, and weight. If the infant is thriving and the answers are reassuring, explain that regurgitation peaks around four months and is gone in nine out of ten babies by the first birthday, and that crying in the first weeks has its own timetable. Use the words spitting up and normal rather than reflux disease. Do not use a questionnaire score or an empiric drug trial to make the diagnosis, and do not prescribe acid suppression for crying or for regurgitation alone. Treat green vomit as a surgical emergency, ask about macrolide antibiotics in the young infant with projectile vomiting, and test the urine when vomiting starts late or comes with faltering growth. When allergy is suspected, run a time-limited formula trial and plan the re-challenge from the start. And when you inherit an infant already on acid suppression without a clear indication, take it away.